

AI/ML DETECTION OF DNS ABUSE AND INFRASTRUCTURE ATTACKS

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Project Proposal Report

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DECLARATION

I hereby declare that this is my own work, and that this proposal has not incorporated, without proper acknowledgment, any material previously submitted for a degree or diploma at any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by any other person, except where explicit acknowledgment is made in the text.

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ABSTRACT

The Domain Name System (DNS) is a foundational pillar of the internet, yet its inherent design vulnerabilities make it a prime target for infrastructure-level attacks that threaten global network stability. The overall purpose of this study is to develop a robust and adaptive framework for the real-time detection of DNS abuse and infrastructure attacks, with a primary focus on Distributed Denial of Service (DDoS) variants such as flooding and amplification attacks. The core research problem is the increasing ineffectiveness of traditional signature-based and static-threshold detection methods against the sophistication and scale of modern threats, compounded by the challenge of acquiring large-scale labeled datasets required for purely supervised machine learning models. The basic design of the study is a hybrid, multi-stage framework that combines lightweight statistical analysis with a suite of advanced machine learning classifiers. The methodology encompasses a systematic pipeline including data preprocessing, feature engineering, and a comparative evaluation of multiple supervised learning models (e.g., Random Forest, XGBoost, SVM). To address the data dependency issue, this research will also investigate the efficacy of a semi-supervised learning paradigm. The framework will be evaluated on the modern dataset, with the anticipated outcome being a highly accurate, computationally efficient, and adaptive detection system capable of identifying malicious traffic patterns with low latency and minimal false positives.

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LIST OF ABBREVIATIONS

Based on the document provided, here is a list of abbreviations used:

- **AI:** Artificial Intelligence
- **API:** Application Programming Interface
- **APT:** Advanced Persistent Threat
- **CPU:** Central Processing Unit
- **DDoS:** Distributed Denial of Service
- **DGA:** Domain Generation Algorithm
- **DNS:** Domain Name System
- **DoH:** DNS over HTTPS
- **DoT:** DNS over TLS
- **FR:** Functional Requirement
- **GeoIP:** Geographical Internet Protocol
- **GUI:** Graphical User Interface
- **ID:** Identifier
- **IP:** Internet Protocol
- **ISP:** Internet Service Provider
- **JSON:** JavaScript Object Notation
- **LSTM:** Long Short-Term Memory
- **ML:** Machine Learning
- **MSSP:** Managed Security Service Provider
- **NFR:** Non-Functional Requirement
- **NIDS:** Network Intrusion Detection System
- **NVMe:** Non-Volatile Memory Express
- **OS:** Operating System
- **RAM:** Random Access Memory

- **RBAC:** Role-Based Access Control
- **RNN:** Recurrent Neural Network
- **ROC-AUC:** Receiver Operating Characteristic - Area Under the Curve
- **SIEM:** Security Information and Event Management
- **SSD:** Solid-State Drive
- **SVM:** Support Vector Machine
- **TC:** Test Case
- **TTL:** Time to Live
- **TTP:** Tactics, Techniques, and Procedures

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1. INTRODUCTION

The Domain Name System (DNS) is a fundamental part of the internet infrastructure, facilitating the resolution of human-readable domain names to IP addresses. However, due to its central role in internet communication, DNS services have become a target for various malicious activities, including DNS abuse and infrastructure attacks. DNS abuse encompasses a wide range of attacks, such as DNS spoofing, cache poisoning, DNS tunneling, and DDoS amplification attacks, which can severely disrupt internet services or enable other forms of cybercrime. These types of attacks exploit the DNS protocol's inherent weaknesses and often go undetected by traditional security measures [1][10].

In recent years, the introduction of DNS over HTTPS (DoH) and DNS over TLS (DoT) protocols has enhanced privacy and security by encrypting DNS queries, preventing eavesdropping and tampering with DNS traffic. However, these encryption methods have inadvertently created new challenges for detecting malicious activity, as traditional detection techniques that rely on inspecting unencrypted traffic patterns can no longer be used. Consequently, there is an increasing need for advanced AI/ML-based solutions that can analyze DNS traffic in real time, even when it is encrypted, to detect anomalies and identify potential attacks [10].

Recent studies have shown the promise of using machine learning (ML) and artificial intelligence (AI) to address these challenges. Various approaches, such as supervised learning with Random Forests, Support Vector Machines (SVM), and deep learning models like Recurrent Neural Networks (RNNs) and Autoencoders, have been employed to detect DNS-based attacks by analyzing traffic patterns and classifying suspicious behavior [1][2]. These methods leverage the power of large datasets and feature extraction techniques to identify patterns indicative of DNS abuse and infrastructure attacks, even in encrypted DNS traffic.

1.1 Background & Literature Survey

Several studies have explored supervised machine learning models for detecting DNS abuse and attacks. For example, Random Forests and Support Vector Machines (SVM) have been applied to identify anomalous DNS traffic and classify it as malicious or benign. These models work by extracting features from DNS traffic, such as query types, response codes, query frequencies, and response times, and then training classifiers to recognize patterns indicative of attacks [4] [5]. However, these traditional methods often face limitations, particularly in the context of encrypted DNS traffic, where certain features are obfuscated by encryption.

Recent advancements in deep learning have also been explored for DNS abuse detection. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which are capable of processing sequential data, have been employed to analyze DNS traffic for patterns over time. These models can capture temporal dependencies and subtle variations in traffic behavior, which are useful for detecting more sophisticated attacks such as DNS tunneling and DGA-based (Domain Generation Algorithm) DNS abuse [6]. Furthermore, autoencoders have been proposed for anomaly detection in DNS traffic, as they can learn compact representations of normal traffic and identify deviations that may indicate malicious activity [7].

But still there are several challenges remaining,

- **Real-time detection:** Most existing models either operate in offline modes or cannot scale effectively for real-time DNS traffic analysis, which is critical for rapid threat mitigation.
- **Encrypted DNS traffic:** The use of DoH and DoT protocols has made traffic analysis more difficult. While traditional methods rely on inspecting the content of DNS queries and responses, the encrypted nature of DoH traffic limits the ability to analyze the data directly. New approaches are needed to focus on traffic patterns and anomaly detection at the network level.
- **Feature selection:** One of the key challenges in applying machine learning to DNS abuse detection is feature selection. Many features, such as timing patterns, response codes, and query frequencies, need to be carefully chosen to maximize detection accuracy. Inaccurate or redundant features can degrade model performance, especially in high-dimensional datasets [8].

In response to these challenges, several techniques have been proposed to enhance DNS abuse detection. Some studies have focused on unsupervised learning techniques like clustering and anomaly detection to identify malicious DNS traffic without requiring labeled datasets. These approaches aim to detect novel attack patterns that have not been previously seen during training [9]. Additionally, hybrid models that combine supervised learning with unsupervised clustering have been explored to improve detection

performance and adaptability to new attack strategies [10]. In recent years, there has been a growing interest in integrating DNS traffic analysis with threat intelligence systems. These systems leverage external threat data to enhance the detection of DNS-based attacks. By correlating DNS traffic with known attack patterns and adversarial tactics, techniques, and procedures (TTPs), these systems can provide more accurate and timely alerts [11].

1.2 Research Gap

Although machine learning (ML) techniques for identifying DNS abuse and infrastructure attacks have advanced, there are still several important research gaps that this project seeks to fill.

Research Gap	Description
Lack of Real-Time Detection in Encrypted DNS Traffic	Most ML models are limited to offline detection or unencrypted DNS traffic. With the rise of DoH and DoT, traditional models become ineffective in analyzing encrypted DNS traffic.
Limited Detection Scope	Existing models focus on specific DNS abuse attacks but fail to address the full range, such as DNS spoofing, cache poisoning, attacks.
Feature Selection and Adaptation	Existing models use static features, which may not be effective for evolving attack vectors. Dynamic feature selection is needed for real-time adaptability.
Scalability and Adaptability of Detection Systems	Many detection systems struggle with scalability and fail to handle large-scale DNS infrastructures, affecting performance and accuracy in larger environments.
Real-World Data Limitations	Most studies rely on small-scale datasets, which may not capture real-world DNS traffic patterns. There is a gap in research using large-scale, real-world datasets.

Table I: Research Gap Explained

As a summary, while machine learning techniques for detecting DNS abuse and infrastructure attacks have shown promise, there are significant gaps in real-time detection, feature selection, and scalability that need to be addressed. This research aims

to fill these gaps by developing a comprehensive AI/ML-based system that can detect a wide range of DNS abuse and infrastructure attacks in real-time and is capable of working with encrypted DNS traffic. Furthermore, the system will utilize dynamic feature selection and be scalable to handle large volumes of DNS traffic [12].

1.3 Research Problem

The increasing use of DNS-over-HTTPS (DoH) and DNS-over-TLS (DoT) has raised significant challenges in the detection of DNS abuse and infrastructure attacks. While these encrypted DNS protocols offer privacy and security benefits by preventing eavesdropping, they inadvertently limit traditional detection methods that rely on inspecting the content of DNS queries and responses. This encryption, combined with sophisticated attack methods like DNS tunneling, DDoS amplification, DNS spoofing, and cache poisoning, makes it difficult for security systems to analyze and detect malicious DNS traffic in real time. Core Research Problem:

“How can AI/ML models be applied to detect and classify DNS abuse and infrastructure attacks in encrypted DNS traffic (DoH/DoT) in real time with a higher accuracy?”

The following specific challenges can be used to further refine this problem:

- Real-time Detection - Develop an AI/ML-based detection system that operates in real time, enabling the identification of malicious DNS traffic as it occurs, without relying on post-event analysis or delay.
- Encrypted DNS Traffic Analysis - Overcome the limitations of traditional DNS abuse detection methods, which rely on unencrypted DNS traffic, by focusing on traffic pattern analysis and anomaly detection in encrypted traffic (DoH/DoT).
- Comprehensive Attack Detection - Identify a broad spectrum of DNS abuse attacks, including DNS tunneling, DDoS amplification, cache poisoning, and DNS spoofing, using machine learning models trained on diverse datasets.
- Scalability and Adaptability - Develop models that are scalable to handle large volumes of DNS traffic in real-world environments and can adapt to evolving attack strategies and new attack vectors.

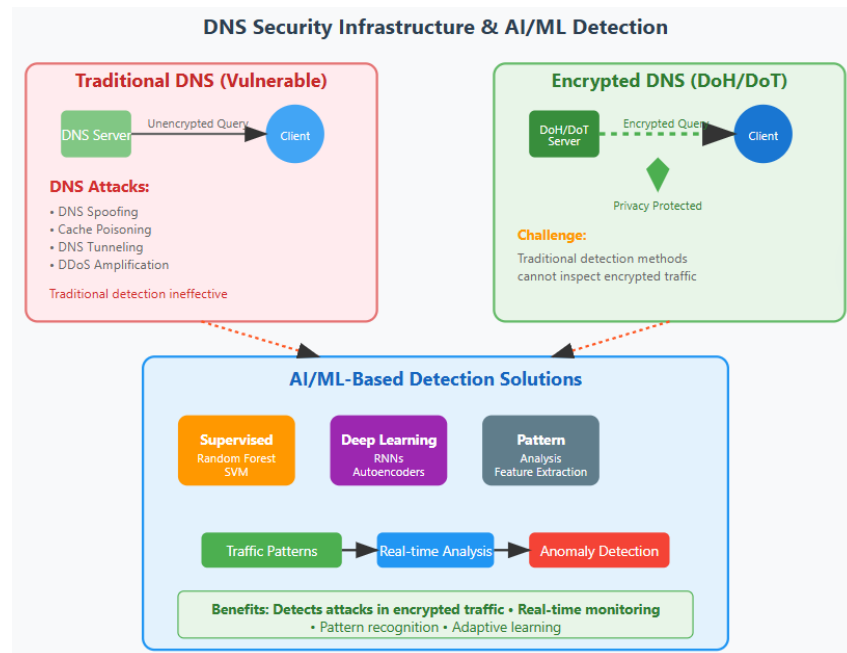


Figure 1: DNS Security & AI ML Detection Diagram

1.4 Objectives

1.4.1 Main Objective

The primary goal of this research is to develop an AI/ML-based detection system capable of identifying a broad range of DNS abuse and infrastructure attacks (including DNS DDoS amplification, DNS spoofing, and cache poisoning) in real-time, even in encrypted DNS traffic (DoH/DoT).

1.4.2 Specific Objectives

Specific Objective	Description
AI/ML Model Development	Implement and train machine learning models (Random Forest, SVM, XGBoost, Deep Learning) for real-time DNS traffic classification into benign or malicious categories.
Encrypted DNS Traffic Analysis	Detect malicious traffic in DNS-over-HTTPS (DoH) and DNS-over-TLS (DoT) by analyzing patterns like query frequencies, response times, query/response lengths, and timing intervals.
Feature Extraction and Selection	Extract key features from DNS traffic (packet size, query types, frequency, unique domain queries) and use dynamic feature selection to adapt to evolving attack strategies.

Real-Time Attack Detection	Design a real-time detection system that monitors DNS traffic continuously, classifies, and alerts about potential attacks to enable immediate countermeasures.
Comprehensive Attack Coverage	Develop a model that detects multiple DNS abuse attack types (DNS tunneling, DDoS amplification, spoofing, cache poisoning, DGA-based attacks).
Scalability and Performance	Evaluate and ensure the scalability of the system to handle high-volume DNS traffic in real-world environments without significant latency or resource overhead.
Comparison with Existing Systems	Compare the performance of the developed model with existing DNS abuse detection systems in terms of accuracy, real-time performance, and scalability.
Integration with Existing Infrastructure	Design the system for easy integration with existing DNS servers or firewalls, providing seamless detection and mitigation of DNS abuse attacks without major network changes.
Evaluation and Optimization	Evaluate the system using benchmark datasets and real-world DNS traffic to measure performance metrics (accuracy, precision, recall, F1-score, ROC-AUC) and optimize minimizing false positives and false negatives.

Table II: Specific Objectives

1.5 Expected Outcomes

- A high-performance AI/ML detection system capable of identifying and classifying DNS abuse and infrastructure attacks in real-time.
- A scalable solution that can be deployed on large DNS infrastructures, ensuring both security and efficiency.
- A comprehensive defense mechanism that covers a wide range of DNS abuse attacks, even in encrypted traffic scenarios.

2. METHODOLOGY

The proposed methodology outlines the steps and techniques used to develop the AI/ML-based system for real-time detection of DNS abuse and infrastructure attacks. This section describes the overall approach, including data collection, model development, evaluation, and integration into existing DNS infrastructures.

2.1 System Architecture and Component Integration.

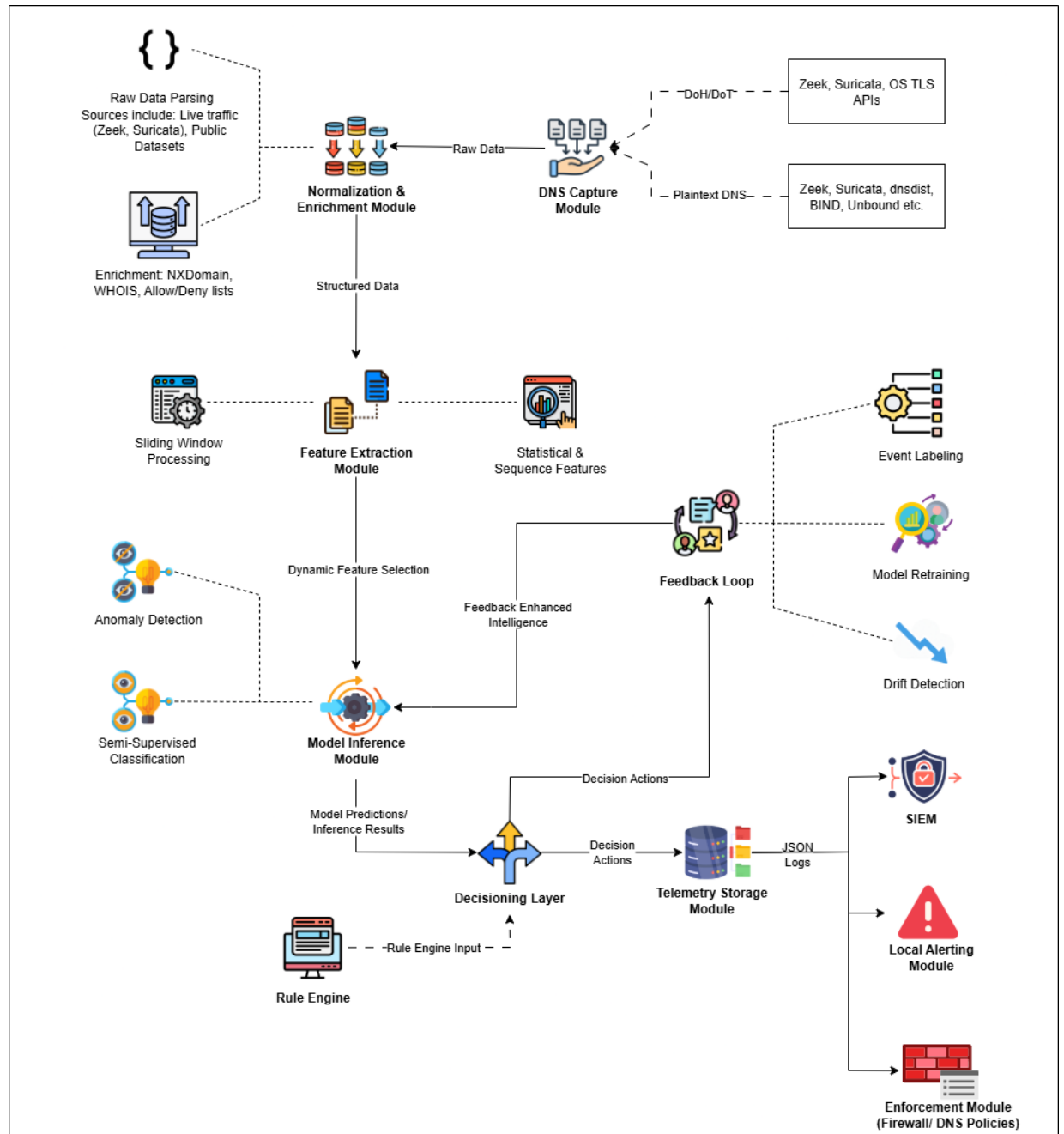


Figure 2: Proposed System Architecture

The architecture of the proposed system is illustrated in the diagram above. It is designed as a closed-loop, adaptive system that not only detects threats but also learns from its outcomes to improve future performance. The integration of the components follows a logical data flow from ingestion to action:

1. **Data Ingestion and Processing:** The process begins at the DNS Capture Module, which collects both plaintext DNS and encrypted DoH/DoT traffic metadata from live network sources (using tools like Zeek/Suricata) and public datasets. This raw data is passed to the Normalization & Enrichment Module, which standardizes the data format and enriches it with contextual information from WHOIS lookups and threat intelligence feeds (Allow/Deny lists).
2. **Feature Engineering:** The resulting structured data flows into the Feature Extraction Module. This component, aided by Sliding Window Processing, calculates statistical and sequence-based features (e.g., query frequency, domain name entropy, response size variance) that are indicative of DNS traffic patterns.
3. **AI-Driven Analysis:** A Dynamic Feature Selection process first identifies the most predictive features, which are then fed into the Model Inference Module. This core analytical engine uses a hybrid approach of Anomaly Detection (for novel threats) and Semi-Supervised Classification (for known threat patterns) to classify the traffic and generate predictions [11].
4. **Decision and Response:** The model's predictions are sent to the Decisioning Layer. This component uses a combination of the model's probabilistic output and predefined rules from the Rule Engine to determine the appropriate action. Based on this decision, it triggers one or more outputs:
 - Logs and telemetry are sent to the Telemetry Storage Module and the SIEM for analysis and correlation.
 - High-priority alerts are dispatched by the Local Alerting Module.
 - Automated defensive actions, such as blocking or rate-limiting, are executed by the Enforcement Module via firewall or DNS policy integration.
5. **Adaptive Feedback Loop:** To ensure the system adapts to evolving threats, the decision outputs are fed into a Feedback Loop. Analysts can perform Event Labeling to confirm alerts. This feedback, combined with automated Drift Detection, triggers the Model Retraining process, which updates the models in the inference engine with enhanced intelligence.

2.2 Description of Tasks and Subtasks

The project will be executed through a series of structured tasks:

Task 1	Setup & Collection	Environment Setup and Data Collection
Subtask 1.1		Deploy and configure network monitoring tools (Zeek, Suricata) on a test network to capture live DNS traffic.
Subtask 1.2		Acquire, validate, and sanitize public datasets containing labeled malicious and benign DNS traffic (e.g., CIC-DDoS2019, CTU-13).
Subtask 1.3		Establish a data processing pipeline for ingesting, normalizing, and storing the collected data.
Task 2	Feature Engineering	Feature Engineering and Selection
Subtask 2.1		Develop scripts to extract key features, including statistical, lexical, and time-series features.
Subtask 2.2		Integrate enrichment APIs for WHOIS and threat intelligence feeds.
Subtask 2.3		Implement and evaluate algorithms for dynamic feature selection to optimize model performance.
Task 3	Model Development	Model Development and Training
Subtask 3.1		Implement and train baseline unsupervised models for Anomaly Detection (e.g., Isolation Forest, Autoencoder).
Subtask 3.2		Implement and train Semi-Supervised Classification models (e.g., Label Propagation, Self-Training Classifiers).
Subtask 3.3		Perform hyperparameter tuning and use k-fold cross-validation to optimize and validate model performance.
Task 4	System Integration	System Integration and Real-Time Implementation
Subtask 4.1		Develop the Decisioning Layer and Rule Engine to translate model outputs into actionable decisions.
Subtask 4.2		Integrate the trained models into a real-time inference pipeline.
Subtask 4.3		Develop API connectors for the Enforcement, Alerting, and SIEM modules.
Task 5	Evaluation & Documentation	Evaluation, Optimization, and Documentation
Subtask 5.1		Evaluate the end-to-end system on a holdout test dataset using metrics such as Accuracy, Precision, Recall, and F1-Score.
Subtask 5.2		Implement and test the complete feedback loop for model retraining and drift detection.
Subtask 5.3		Document the system architecture, code, and experimental results in the final research paper.

Table III: Task Overview

2.3 Data Collection Methods and Tools

Data Sources:

- **Live Network Traffic:** Captured from a controlled network environment using Network Security Monitors to provide realistic, unlabeled data.
- **Public Datasets:** Academic and industry datasets (e.g., CIC-DDoS2019, CTU-13, DoH-CIRA) will provide ground truth for training and testing supervised components.

Tools and Technologies:

- **Data Capture:** Zeek, Suricata.
- **Programming & Data Manipulation:** Python, Pandas, NumPy.
- **Machine Learning Libraries:** Scikit-learn, TensorFlow, PyTorch.
- **Data Storage:** Elasticsearch or a similar time-series database for telemetry.

2.4 Expected Timeframes for Tasks

The research is projected to be completed over a 7-month period, as outlined below:

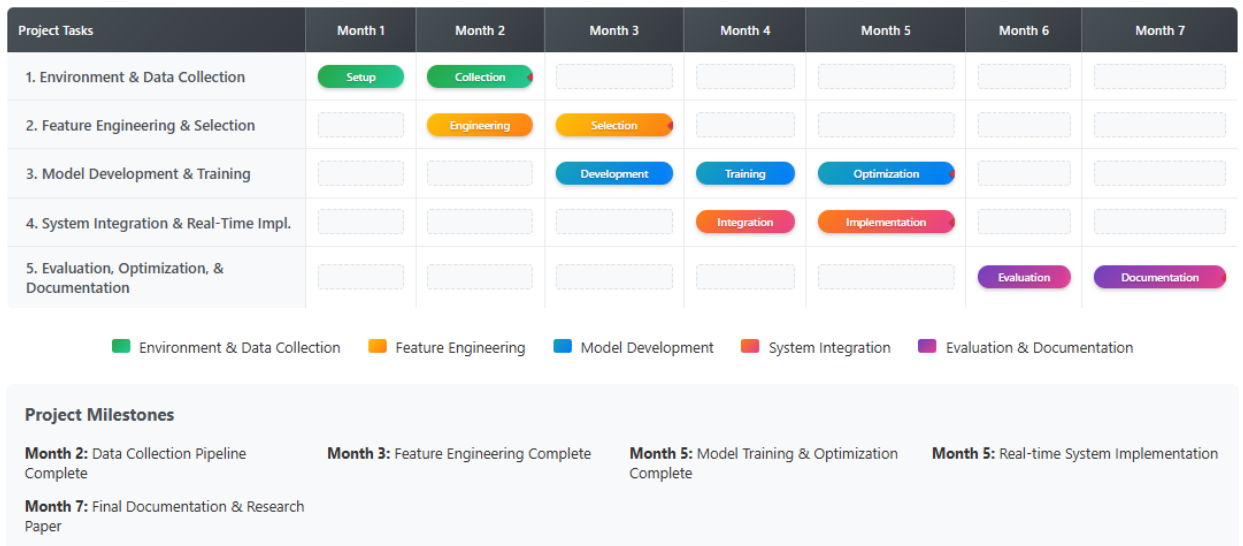


Figure 3: Project Timeline

2.5 Conclusion and Real-World Applications of the Research

The proposed methodology provides a comprehensive and systematic plan for developing a next-generation DNS threat detection system. By integrating advanced AI/ML techniques with a real-time, adaptive architecture, this research directly addresses critical gaps in existing solutions, particularly in handling encrypted traffic and adapting to evolving attack vectors. The successful implementation of this system will result in a robust, scalable, and intelligent defense mechanism capable of protecting critical DNS infrastructure from a wide range of abuses and attacks [10].

The outcomes of this research have significant practical applications across various sectors:

- **Internet Service Providers (ISPs):** To protect their infrastructure and customers from botnet command-and-control (C2) communication, DDoS attacks, and phishing campaigns at a network-wide scale.
- **Enterprise Security:** For corporations to defend against advanced persistent threats (APTs), data exfiltration via DNS tunneling, and malware infections within their internal networks.
- **Managed Security Service Providers (MSSPs):** To offer advanced DNS security and threat intelligence as a service to their clients.
- **Public DNS Resolvers:** To enhance the security and integrity of their services, providing a safer browsing experience for millions of users by filtering out malicious domains in real time.

2.6 Anticipated Conclusion

The anticipated outcome of this research is a highly effective and computationally efficient framework for detecting DNS abuse and infrastructure attacks. It is expected that the supervised machine learning models, particularly ensemble methods like Random Forest and XGBoost, will achieve a high detection accuracy (over 98%) on the CIC-DDoS2019 dataset, significantly outperforming the statistical baseline model [5]. Furthermore, it is anticipated that the semi-supervised learning approach will demonstrate its value by successfully leveraging unlabeled data to maintain high accuracy, proving it to be a viable solution for environments where labeled data is scarce. The systematic feature selection process is expected to identify a smaller, more potent subset of features that can reduce model complexity and inference time without a significant loss in performance [4][6].

The real-world application of this project is a more resilient DNS infrastructure. The final model can be integrated into a real-time Network Intrusion Detection System (NIDS) to provide security analysts with timely and accurate alerts, enabling them to mitigate attacks before they cause significant service disruption. This research will contribute to a validated, comparative analysis of modern detection techniques, offering a practical blueprint for enhancing the security of critical internet services [6][7].

3. PROJECT REQUIREMENTS

The development of an AI/ML-based system for detecting DNS abuse and infrastructure attacks requires clearly defined requirements to ensure the system is effective, scalable, and suitable for real-world deployment. These requirements are categorized into functional, non-functional, and user requirements, along with tentative use cases and test cases.

3.1 Functional Requirements

Functional requirements define the specific behaviors and functions of the system, what it must do.

ID	Requirement	Description
FR-01	DNS Traffic Ingestion	The system must be able to capture and ingest both plaintext DNS traffic (from sources like BIND, Zeek) and encrypted DNS traffic metadata (DoH/DoT).
FR-02	Data Normalization	All ingested data from various sources must be parsed and transformed into a standardized, structured format for consistent processing.
FR-03	Data Enrichment	The system must enrich normalized data with external context, including WHOIS information, threat intelligence feeds (allow/deny lists), and GeoIP data.
FR-04	Feature Extraction	The system must extract statistical, lexical, and time-series features from the DNS traffic. This includes query frequency, response size, TTL values, and domain name entropy.

FR-05	AI-Based Threat Detection	The system must use machine learning models (semi-supervised and unsupervised) to analyze the extracted features and classify DNS traffic as either benign or malicious.
FR-06	Threat Classification	Malicious traffic must be classified into specific attack categories where possible (e.g., DNS Tunneling, DDoS Amplification, DGA).
FR-07	Real-Time Alerting	The system must generate real-time alerts for any DNS traffic classified as malicious. Alerts must be prioritized based on severity.
FR-08	Alert Dashboard	The system must provide a user interface (dashboard) to display key metrics, including total traffic analyzed, active alerts, and threat trends over time.
FR-09	Alert Investigation	The user interface must allow a security analyst to drill down into specific alerts to view detailed information, including traffic logs, features, and the model's prediction confidence.
FR-10	Automated Enforcement	The system must be able to send automated mitigation commands (e.g., Block IP, Rate-Limit) to an integrated enforcement point (Firewall/DNS Server) based on predefined rules.
FR-11	System Logging	All system activities, including data ingestion, model predictions, alerts, and enforcement actions, must be logged for auditing and forensic analysis.
FR-12	Feedback Mechanism	The system must allow an analyst to provide feedback on alerts (e.g., mark as true/false positive), which will be used for model retraining.

Table IV: Funtional Requirements

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system and how well it performs its functions.

ID	Requirement	Description
NFR-01	Performance	The system must process incoming DNS traffic with a latency of less than 200 milliseconds from ingestion to decision. It must be capable of handling a baseline of 10,000 DNS queries per second.
NFR-02	Scalability	The system's architecture must be scalable to handle a 3x increase in traffic volume without significant degradation in performance.
NFR-03	Reliability	The system must have an uptime of 99.9% and include mechanisms for fault tolerance and graceful degradation.
NFR-04	Security	Access to the system's dashboard and APIs must be secured via role-based access control (RBAC). All sensitive data (logs, model configurations) must be encrypted at rest.
NFR-05	Maintainability	The system must be modular to allow for independent updates of components (e.g., adding a new model). It must provide comprehensive logging for debugging and monitoring.
NFR-06	Usability	The user dashboard must be intuitive and provide clear visualizations, enabling a security analyst to identify and investigate threats with minimal training.

Table V: Non-Functional Requirements

3.3 User Requirements.

User requirements are written from the perspective of the end-users.

Primary Users:

- Security Analyst: Responsible for monitoring network security, investigating threats, and responding to incidents.
- Network Administrator: Responsible for maintaining the health and performance of the network infrastructure.

User	Requirement Statement
Security Analyst	- As a Security Analyst, I need a centralized dashboard to get a real-time overview of DNS-based threats so that I can quickly assess the security posture of the network. - As a Security Analyst, I need to receive immediate, high-priority alerts for critical threats like DNS tunneling so that I can begin incident response without delay. - As a Security Analyst, I need to be able to investigate an alert and see all the contextual data (source IP, domain, query type, model score) in one place so that I can make an accurate determination of the threat. - As a Security Analyst, I need to provide feedback on the accuracy of alerts so that the system can learn and reduce false positives over time.
Network Admin	- As a Network Administrator, I need the system to integrate with our existing firewall so that it can automatically block malicious IPs without manual intervention. - As a Network Administrator, I need to be able to monitor the performance of the detection system itself to ensure it is not negatively impacting network latency.

Table VI: User Requirements Table

3.4 System requirements

Development Environment:

- OS: Linux (Ubuntu 20.04+ recommended)
- Programming Language: Python 3.8+
- Key Libraries: Scikit-learn, TensorFlow/PyTorch, Pandas, Zeek/Suricata for data capture.
- Hardware: 16GB RAM, 8-core CPU, 512GB SSD.

Production Deployment Environment (per node):

- OS: Linux (Ubuntu 20.04+ or similar server distro)
- Hardware: 32GB+ RAM, 16-core CPU, 1TB+ high-speed storage (SSD/NVMe).
- Network: 10Gbps network interface card.

3.5 Tentative Use Case

Use Case 1: Real-time DNS Attack Detection

- **Actor:** Network Security Analyst
- **Description:** The system monitors network traffic in real-time, detects a DNS DDoS attack, and alerts the analyst.
- **Flow:**
 1. Malicious DNS traffic (e.g., a high-rate amplification attack) begins targeting the network.
 2. The system's DNS Capture Module ingests the traffic.
 3. The Feature Extraction Module processes the traffic and generates feature vectors.
 4. The Model Inference Module classifies the traffic as malicious.
 5. The Decisioning Layer confirms the attack.
 6. The system sends a high-priority alert to the SIEM and the Local Alerting Module.
 7. The Network Security Analyst receives the alert and begins the incident response process.

Use Case 2: Forensic Analysis of a Past Attack

- **Actor:** Network Security Analyst
- **Description:** After an attack has been mitigated, the analyst uses the system's logs to investigate the incident.
- **Flow:**
 1. The analyst queries the Telemetry Storage Module for all events related to a specific timeframe or IP address.
 2. The system retrieves and displays the structured JSON logs for the detected attack.
 3. The logs provide details on the traffic characteristics, model predictions, and decision actions taken.
 4. The analyst uses this information to understand the attack vector and prepare a post-mortem report.

3.6 Tentative Test Cases for System Evaluation

The following test cases will be used to validate the functionality and performance of the final system.

Test Case ID	Description	Input Data	Expected Outcome
TC-01	Benign Traffic Validation	A subset of the CIC-DDoS2019 dataset containing only normal, benign DNS traffic.	The system should classify >99% of the traffic as benign, generating a minimal number of false positive alerts.
TC-02	High-Rate DNS Flood Detection	A subset of the dataset containing a high-volume DNS flooding attack.	The system should detect the attack with high recall (>99%) and generate a corresponding alert.
TC-03	DNS Amplification Detection	A subset of the dataset containing a DNS amplification attack.	The system should detect the attack with high recall (>99%) and generate a corresponding alert.
TC-04	Low-Rate Attack Sensitivity	A mixed dataset with benign traffic and a simulated low-and-slow DNS attack.	The system should successfully detect the subtle attack pattern, demonstrating high sensitivity (high recall).
TC-05	Performance Under Load	A very large dataset simulating high traffic volume.	The system should process the data and generate classifications within an acceptable real-time latency threshold (e.g., sub-second per flow).

Table VII: Tentative Test Cases

3.7 Wireframes

This view provides a high-level overview of the network's DNS security posture.

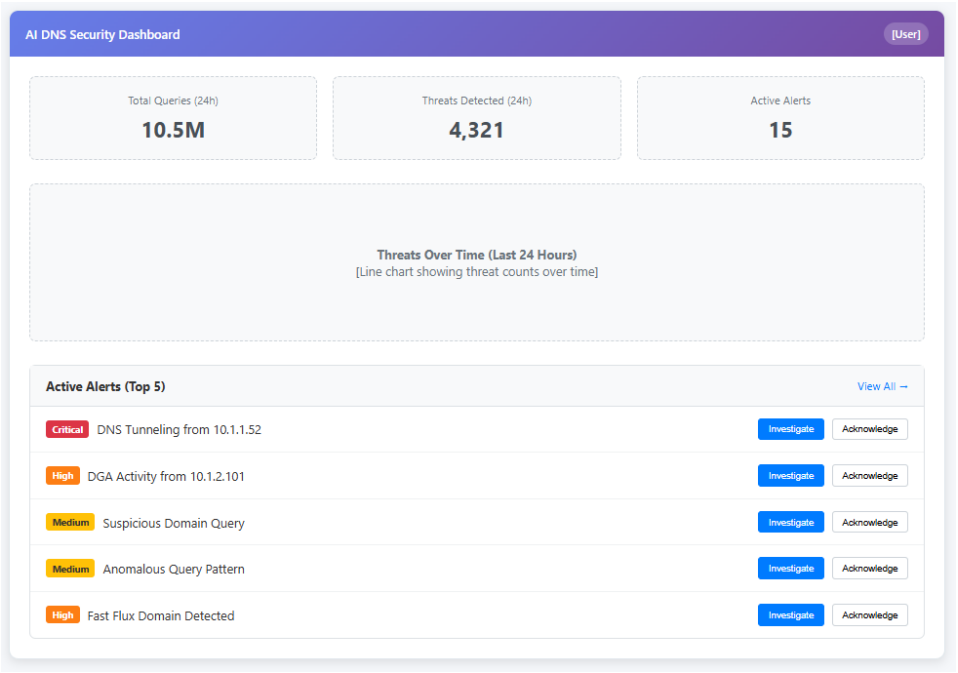


Figure 4: Wireframe 1: Main Dashboard

This view allows an analyst to drill down into a specific alert for detailed analysis.

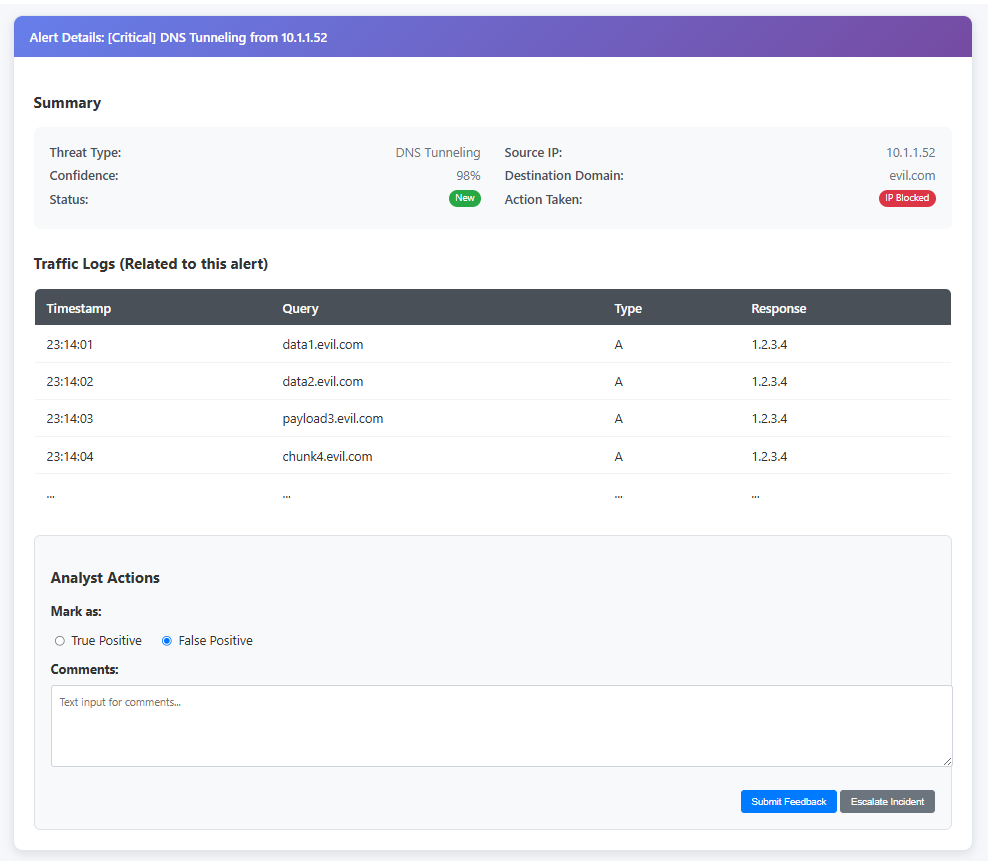


Figure 5: Wireframe 2: Alert Investigation View

As this research component focuses on the backend AI/ML detection engine, traditional graphical user interface (GUI) wireframes are not directly applicable. The primary "interface" for the user (a security analyst) is the structured alert data that the system generates and sends to a Security Information and Event Management (SIEM) platform or a logging system.

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